Exp 7: Design a controller with servlet that provides the interaction with application developed in experiment 1 (registration form)and the database created in experiment 5.

Required directory structure:

In tomcat folder

→ web apps

→ create ur application folder (e.g. task 7)

→ create WEB-INF folder

(inside this folder create)

→ classes
lib
web.xml

→ create register.html here

- in classes folder : java source file and javax.servlet jar file is stored.
- In lib folder : mysql connector jar file is stored.

- In mysql create a database registrationdb(last 3digits of ur roll no), and a table logindetails consisting of username, password, email and contactno

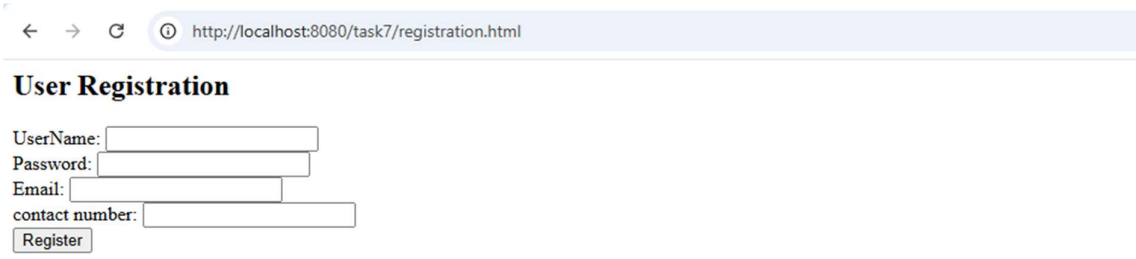
- While compiling java source code include mysql connector jar file and javax.servlet jar file

Compiling :

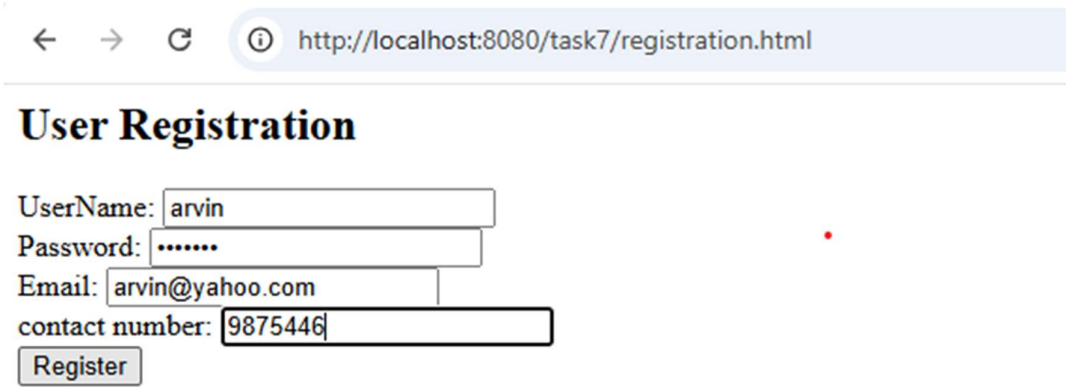
```
C:\Program Files\Apache Software Foundation\Tomcat 9.0\webapps\task7\WEB-INF\classes>javac -cp "javax.servlet-api-3.0.1.
jar;C:\Program Files\Apache Software Foundation\Tomcat 9.0\webapps\task7\WEB-INF\lib\mysql-connector-j-9.6.0.jar;" Regis
terServlet.java

C:\Program Files\Apache Software Foundation\Tomcat 9.0\webapps\task7\WEB-INF\classes>
```

- Select startup.batch file to start tomcat server

Output :



Enter the details and click register



http://localhost:8080/task7/RegisterServlet

You are successfully registered!

Means login details are stored in the database , now check the db

```
mysql> USE registrationdb
Database changed
mysql> select * from logindetails;
+----------+----------+------------------+-----------+
| username | password | email            | contactno |
+----------+----------+------------------+-----------+
| anitha   | 1234     | anitha@gmail.com | 12345     |
+----------+----------+------------------+-----------+
1 row in set (0.00 sec)

mysql> select * from logindetails;
+----------+----------+------------------+-----------+
| username | password | email            | contactno |
+----------+----------+------------------+-----------+
| anitha   | 1234     | anitha@gmail.com | 12345     |
| arvin    | arvin@1  | arvin@yahoo.com  | 9875446   |
+----------+----------+------------------+-----------+
2 rows in set (0.00 sec)

mysql>
```

Programs :

registration.html

```
<html>
<body>
   <h2>User Registration</h2>
   <form method="post" action="RegisterServlet">
      UserName: <input type="text" name="name"><br>
            Password: <input type="password"
name="password"><br>
Email: <input type="text" name="email"><br>
```

```
contact number: <input type="text" name="contact"><br>
        <input type="submit" value="Register">
    </form>
</body>
</html>
```

RegisterServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.sql.*;

public class RegisterServlet extends HttpServlet
{
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException
{
        response.setContentType("text/html;charset=UTF-8");
        PrintWriter pw = response.getWriter();

        String name = request.getParameter("name");
String password = request.getParameter("password");
        String email = request.getParameter("email");
        String contactno = request.getParameter("contact");

String url = "jdbc:mysql://localhost:3306/registrationdb5_ _ ";  //
JDBC URL
        String username1 = "root";  // MySQL username
        String pwd = " ";  // MySQL password

try {
            // Load the JDBC driver
            Class.forName("com.mysql.jdbc.Driver");
```

```
            // Establish connection (replace with your DB details)
            Connection con = DriverManager.getConnection(url,
username1, pwd);

            // Insert data into the database
            PreparedStatement ps = con.prepareStatement("INSERT
INTO logindetails(username,password,email,contactno) VALUES (?,
?, ?,?)");
            ps.setString(1, name);
            ps.setString(2, password);
            ps.setString(3, email);
            ps.setString(4, contactno);

            int rowsAffected = ps.executeUpdate();
            if (rowsAffected > 0) {
                pw.println("You are successfully registered!");
            } else {
                pw.println("Registration failed.");
            }
            con.close();
        } catch (Exception se) {
            se.printStackTrace();
            pw.println("An error occurred: " + se.getMessage());
        }
    }
}
```

web.xml

```
<web-app>
    <servlet>
        <servlet-name>RegisterServlet</servlet-name>
        <servlet-class>RegisterServlet</servlet-class>
    </servlet>
    <servlet-mapping>
```

```
        <servlet-name>RegisterServlet</servlet-name>
        <url-pattern>/RegisterServlet</url-pattern>
    </servlet-mapping>
</web-app>
```